Session routing by delta activation

SessionControl · implemented write → route → answer architecture

(A) Write and store session-specific deltas

Session input

Full answer-bearing session

Original question q_s

Supplied answer a_s

Repeat independently for each session

Session-specific memory write

AlphaEdit grounded optimization

FFN down-projections · layers 13–15

KEY Q: q_s A:

TARGET a_s + EOS

$$\Delta W_s = W_{\text{edited},\,s} - W_{\text{base}}$$

Restore base weights after every write

Independent session delta bank

ΔW_1 ΔW_2 … ΔW_N

Store each delta separately; keep the base model fixed

(B) Route by activation, then generate the answer

Read one stored delta at a time

Original question q

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Q: {question}
A:
```

Question only at recall

Base LLM forward pass

W_{base}

Capture FFN input activations

$k_\ell(q)$ at the final key token

$k_\ell(q)$

Delta activation router

Score every stored delta using the same query activations

$$E_s(q) = \frac{\sum_{\ell \in L} \|\Delta W_{s,\ell}\, k_\ell(q)\|_2^2}{\sum_{\ell \in L} \|\Delta W_{s,\ell}\|_F^2}$$

Select $s^* = \arg\max_s E_s(q)$

Equal scores: fixed lexicographic session-ID tie break

ΔW_{s^*}

Answer LLM

$W_{\text{base}} + \Delta W_{s^*}$

Activate the selected delta

Greedy generation

EOS / next-turn stop

Generated answer

From the selected memory

Same original question q

$L = \{13, 14, 15\}$ · normalized delta activation energy

Restore base weights after answer generation

Exact-question experiment: recall uses the question supplied at write time; one Q/A is supervised per session.

Our session-memory protocol and activation routing

Existing inputs, AlphaEdit, and LLM computation